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# Split the dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)

# Create KNN model
model = KNeighborsClassifier(n_neighbors=5)

# Train the model
model.fit(X_train, y_train)

# Predict
y_pred = model.predict(X_test)

# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", round(accuracy, 4))

# Generate confusion matrix
conf_matrix = confusion_matrix(y_test, y_pred)

# Plot confusion matrix
plt.figure(figsize=(8, 6))
sns.heatmap(
    conf_matrix,
    annot=True,
    fmt='d',
    cmap='summer',
    xticklabels=wine.target_names,
    yticklabels=wine.target_names
)

plt.xlabel("Predicted Label")
plt.ylabel("True Label")
plt.title("Confusion Matrix")

plt.show()

```

Accuracy: 0.7837

